Linux Fleet Automation System Architecture Reference

Centralized logging, host visibility, and identity-aware automation for the managed Linux estate — pipeline design, role execution order, and operational playbooks.

September 2026

Overview

This document is a visual reference for the automation stack that provides centralized log collection, host inventory reporting, and a unified operational dashboard across the managed Red Hat / Linux fleet. It covers four views: the overall system layout, the internal pipeline design of the collector host, and the execution order of the two primary automation playbooks that build and maintain the stack.

The platform is built as reusable, idempotent configuration-management roles, organized so that each capability (log ingestion, dashboard rendering, health checks, alerting) can be deployed, tested, and rolled back independently. It is currently being validated in a controlled test environment ahead of promotion into the production estate.

What's inside

1. System overview — how log clients, the identity realm, the collector host, dashboards, and email alerting fit together

2. Log collector internals — the two pipelines that run on the collector host: centralized logging and the fleet dashboard

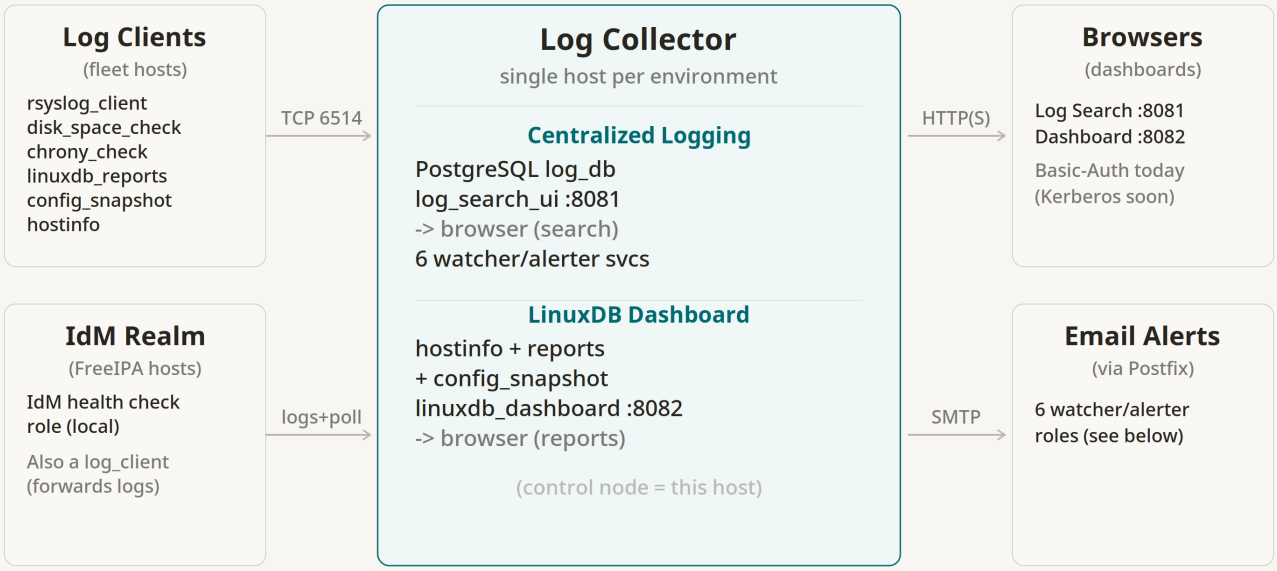
3. Centralized logging automation — the role execution order used to build the logging pipeline end to end

4. Dashboard automation — the orchestration used to build the fleet inventory and reporting dashboard

5. Operational playbook reference — the supporting playbooks used for day-to-day operations and maintenance

1. System Overview

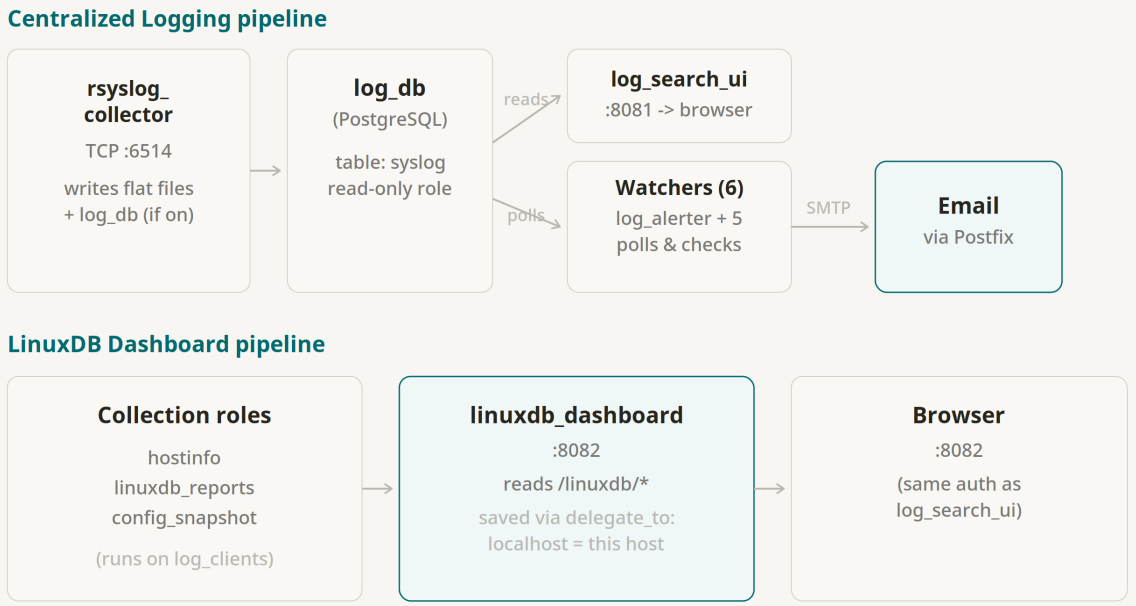
End-to-end view of how fleet hosts, the identity realm, the collector host, browser-based dashboards, and email alerting connect.



- Every managed host forwards logs over TLS to a single collector host and authenticates against a shared identity realm.
- The collector host stores structured events in a database and flat files, and serves two independent web dashboards.
- Alert conditions (service failures, disk thresholds, sync issues) are emailed out through a relay on the collector host.

2. Inside the Collector Host — Two Pipelines

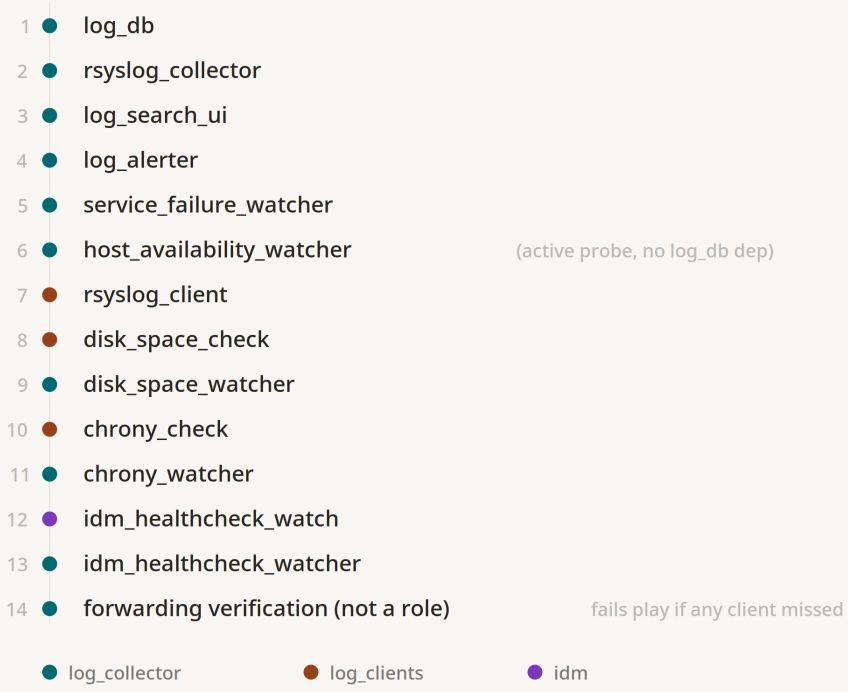
The collector host runs two independent pipelines side by side, sharing the same host but not the same data path.



- Centralized Logging: a syslog listener accepts events on a dedicated port, writes them to a database and flat files, and serves a log search interface over HTTP(S). A set of watcher roles monitor for failures and send email alerts.
- Fleet Dashboard: a set of local collection roles gather host inventory, report data, and configuration snapshots directly on the collector host (no database), and serve them through a second, independent web interface.
- Both interfaces are lightweight, dependency-free web applications; a shared authentication layer is being upgraded to identity-realm-based single sign-on with corporate-issued TLS certificates.

3. Centralized Logging — Automation Role Order

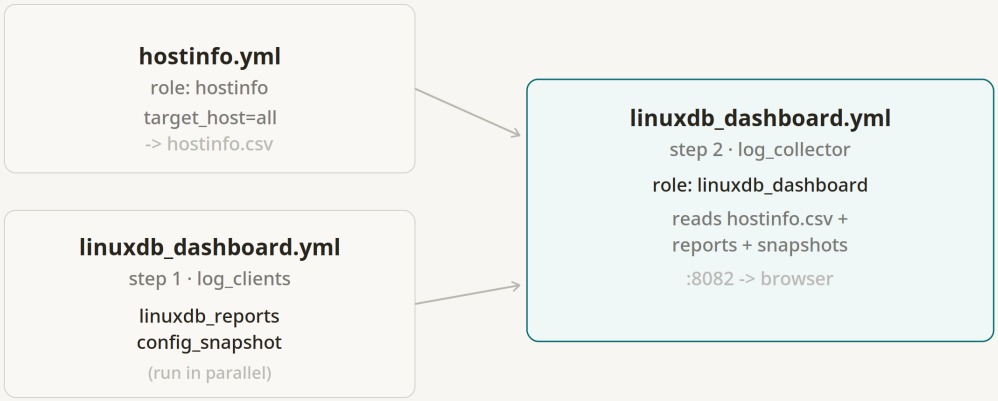
The centralized logging automation applies roles in a fixed, dependency-aware order across three host groups: the collector, log-forwarding clients, and the identity realm.



- Collector-side roles provision the database, syslog listener, log search interface, and the alerting/watcher roles, in that order.
- Client-side roles configure each fleet host to forward logs securely to the collector.
- Identity-realm roles ensure authentication and time synchronization are consistent across the estate before the pipeline goes live.

4. Fleet Dashboard — Automation Orchestration

The dashboard automation gathers host inventory and reporting data locally on the collector host and renders it through a dedicated web interface — no external database required.



- Host inventory, reporting, and configuration-snapshot roles run first and write structured data to the local filesystem.
- The dashboard role then reads that data directly to render the fleet-wide view — keeping the design simple and dependency-free.

Operational Playbook Reference

Supporting playbooks used alongside the two primary automation flows for day-to-day operations, health checks, and maintenance tasks.

Playbook	Purpose	Target group
idm_healthcheck	Validates identity realm services and replication health	idm
system_update / update_system	Applies patches through the package manager, with pre/post checks	all / fleet-wide
system_poweroff / poweroff_all	Controlled shutdown of a single host or the full fleet	targeted / all
silence_setroubleshoot	Suppresses noisy SELinux troubleshooting notifications	log_clients
sudo_audit_report	Reports on sudo usage and privilege escalation events	all
dnf_history	Captures package transaction history for audit purposes	all
gather_critical_alerts	Collects and summarizes critical alert conditions fleet-wide	log_collector
collect_ilo	Gathers out-of-band (iLO/BMC) health data from physical hosts	physical hosts
deploy_chronyd / copy_idm_chrony_conf	Deploys and standardizes time synchronization configuration	all / idm
idm_copy_gssapi_delegation	Applies GSSAPI delegation configuration for identity-aware services	idm
push_script_and_run	Ad hoc utility to push and execute a script across target hosts	variable
why_running	Diagnostic playbook to identify why a service or process is active	targeted
scom	Linux-side integration checks for enterprise monitoring	all

Note: service_failure_alert has been retired and superseded by the watcher roles in the centralized logging pipeline.